

Math 112 — Mathematics of Finance

Study summary, organized by concept. Source: *Mathematics of Finance (Introduction to Actuarial Mathematics)*, P.C. Nwokolo (Immaculate Publications).

This is the whole course compressed into the parts that actually get tested. Read the **Fast Track** first — those five ideas generate most of the exam. Every worked example from the book is kept; almost all of the repetition is gone.

Notation used throughout

- $R = 1 + \frac{r}{100}$ — the growth factor per period. r is the rate **per period** (already divided by frequency).
 - $r\%$ means $\frac{r}{100}$ (a decimal). So “2%” = 0.02.
 - $s_{\bar{n}|r}$ — **accumulation** factor of an annuity (future value of 1 per period).
 - $a_{\bar{n}|r}$ (written $\alpha_{\bar{n}|r}$ in the book) — **present value** factor of an annuity.
 - p = principal / present value, A = amount / final value, n = number of periods, t = time in years, d = periodic payment or periodic interest.
-

Fast Track — the five things that generate the exam

Ranked by how often they appear and how many marks they unlock.

1. Compound interest engine: $A = pR^n$

Everything after Chapter 1 is a variation of this.

$$A = p(1 + \frac{r}{100})^n = pR^n \quad p = AR^{-n}$$

- r is the rate **per conversion period**, n is the **number of periods** — not years. If 8% is “compounded quarterly for 5 years”, then $r = 8/4 = 2$ and $n = 5 \times 4 = 20$.
- Present value (discounting) is the same equation solved backwards: $p = AR^{-n}$.
- Solve for n with logs: $\log(A/p) = n \log R$.

2. The two annuity factors: $s_{\bar{n}|r}$ and $a_{\bar{n}|r}$

$$s_{\bar{n}|r} = \frac{R^n - 1}{r\%} = \frac{(1 + r\%)^n - 1}{r\%} \quad a_{\bar{n}|r} = \frac{1 - R^{-n}}{r\%}$$

- **Future value** of a level annuity (deposit d each period): $A = d s_{\bar{n}|r}$.
- **Present value** of a level annuity (payment d each period): $p = d a_{\bar{n}|r}$.
- These two factors drive annuities, amortization, sinking funds, **and** bonds. Learn them cold.

3. Present value / discounting

Money later is worth less now. Two different worlds — do not mix them:

- **Compound world:** $p = AR^{-n}$.

- **Simple-interest world (short-term):** true discount $p = \frac{A}{1 + rt/100}$; banker's discount $p = A(1 - dt/100)$.

4. Amortization vs. sinking fund (debt extinction)

Two ways to kill a debt. They are examined side by side.

- **Amortization:** each payment covers the period's interest **plus** part of the principal. Payment $d = p/a_{\bar{n}|r}$. Outstanding balance = PV of the payments still to come.
- **Sinking fund:** you pay the lender **interest only** each period, and separately deposit into a fund that grows to the principal: deposit $d = A/s_{\bar{n}|r}$.

5. Bond valuation (purchase price)

$$V_B = p + I = AR^{-n} + d a_{\bar{n}|r} \quad \Leftrightarrow \quad V_B = A + (d - Ar\%) a_{\bar{n}|r}$$

- A = redemption value, d = bond interest per period = (bond rate per period) $\times$ face value, r = **yield** per period, n = number of coupon periods.
- If bond rate $>$ yield $\rightarrow$ price is a **premium** (above redemption). If bond rate $<$ yield $\rightarrow$ **discount** (below).

Chapter 1 — Interest, Present Value, and Discounting

1.1 Simple interest

Interest accrues only on the original principal.

$$i = \frac{prt}{100} \quad A = p(1 + \frac{rt}{100})$$

i = interest, p = principal, r = rate % per year, t = time in **years**, A = amount.

Time from days. Two conventions — the exam loves to switch between them:

$$t = \frac{\text{days}}{365} \text{ (exact / actual)} \quad t = \frac{\text{days}}{360} \text{ (ordinary / banker's)}$$

1.2 Discounting

Finding what a future amount is worth **now**.

Sales / trade discount (a price reduction, not time value):

$$S = S_0(1 - \frac{r}{100})$$

True discount — interest on the *present value*, at simple interest:

$$D = \frac{prt}{100}, \quad p = \frac{A}{1 + rt/100}, \quad D = A - p$$

Banker's discount — interest charged on the *maturity value* A (so it's always a bit larger than true discount):

$$D = \frac{A dt}{100}, \quad p = A(1 - \frac{dt}{100}), \quad A = \frac{p}{1 - dt/100}$$

Here d is the **discount rate**. Relationship between a discount rate d and interest rate r (with $m = t/100$):

$$d = \frac{r}{1 + rm}, \quad r = \frac{d}{1 - dm}$$

Promissory note. A note has a face value, a date, a term, and (sometimes) a stated interest rate. To find proceeds when it's discounted early: (1) find the **maturity value** A (face + interest to maturity), (2) apply **banker's discount** over the time from the discount date to the maturity date.

1.3 Compound interest

Interest earns interest.

$$A = pR^n = p(1 + \frac{r}{100})^n, \quad R = 1 + \frac{r}{100}$$

$$\log \frac{A}{p} = n \log R \quad (\text{solve for } n), \quad i = A - p = p(R^n - 1)$$

Growing annual investment — principal p plus a fixed amount d added at the end of each year:

$$A = pR^n + d(\frac{R^n - 1}{R - 1})$$

With withdrawals of d each period: $A = pR^n - d s_{\bar{n}|r}$.

Present value / discount value under compounding:

$$p = AR^{-n}, \quad D = p(R^n - 1) = A(1 - R^{-n})$$

Nominal vs. effective (APR) rate. A nominal rate compounded m times has a larger *effective* annual rate:

$$e = 100(R^n - 1) \quad \text{per year, with } R \text{ and } n \text{ the per-period figures}$$

Worked example (rates). 16% compounded quarterly $\rightarrow$ per period $r = 4\%$, $n = 4$. Effective rate $= 100((1.04)^4 - 1) = 100(1.16986 - 1) \approx 17\%$. So “16% quarterly” actually earns about **17% a year**.

Why this trips people up: forgetting to convert the rate and periods to *per-conversion-period* before using R^n . “Per annum, compounded monthly” means divide the rate by 12 and multiply the years by 12 — every time.

Chapter 2 — Annuities

An annuity is a **sequence of equal payments** at equal time intervals. Classified by timing (ordinary = end of period; due = start) and by term.

2.1–2.3 Amount (future value) of an annuity

At simple interest the deposits form an arithmetic-style sum. **At compound interest** (the case that matters):

$$A = d(\frac{R^n - 1}{R - 1}) = d s_{\bar{n}|r}, \quad s_{\bar{n}|r} = \frac{R^n - 1}{r\%} = \frac{(1 + r\%)^n - 1}{r\%}$$

d = deposit each period, accumulated to just after the last deposit.

2.4 Present value of an annuity

$$p = d \left(\frac{1 - R^{-n}}{R - 1} \right) = d a_{\bar{n}|r} = AR^{-n}, \quad a_{\bar{n}|r} = \frac{1 - R^{-n}}{r\%}$$

This is the lump sum **today** equivalent to receiving d per period for n periods.

2.5 Perpetuity

Payments that never end. Since $R^{-n} \rightarrow 0$, the present value collapses to:

$$p = \frac{d}{R - 1} = \frac{100d}{r}$$

2.6 Capitalization / capitalized cost

The total endowment needed to buy an asset **and** replace it forever.

- **Replacement every period ($n = 1$ case):**

$$C = p + \frac{d}{R - 1} = p + \frac{100d}{r}$$

- **Replacement every n periods** (renewal cost D set aside via a sinking fund):

$$C = p + \frac{D}{R - 1} \cdot \frac{1}{s_{\bar{n}|r}}, \quad \text{equivalently} \quad C = \frac{p}{R - 1} \cdot \frac{1}{a_{\bar{n}|r}}$$

Why this trips people up: confusing $s_{\bar{n}|r}$ (future value, grows the deposits) with $a_{\bar{n}|r}$ (present value, discounts the payments). Rule of thumb: **saving up** $\rightarrow s$; **paying off** $\rightarrow a$.

Chapter 3 — Debt Extinction: Amortization and Sinking Funds

3.1 Amortization

Repay a loan p with n equal payments; each payment is part interest, part principal.

$$p = d a_{\bar{n}|r} \quad \Rightarrow \quad d = \frac{p}{a_{\bar{n}|r}}$$

- **Interest portion** of a payment = (rate) $\times$ (outstanding balance at start of period).
- **Principal portion** = payment – interest portion.
- **Outstanding principal** at any point = present value of the payments **still remaining** = $d a_{\bar{k}|r}$ for k payments left.

Buyer's / seller's equity. At any moment:

buyer's equity + seller's equity = selling price

Buyer's equity = what the buyer has paid off (principal reduced); seller's equity = outstanding balance.

Add-on interest (a *different*, more expensive scheme):

total owed = principal + simple interest on the whole principal, payment = $\frac{\text{total}}{n}$

Charging simple interest on the full principal the whole term makes the *effective* rate much higher than the quoted rate.

3.2 Sinking fund

Pay the lender **interest only** each period; separately, build a fund that reaches the principal A by maturity.

$$\text{periodic deposit } d = \frac{A}{s_{\bar{n}|r}}, \quad \text{periodic cost} = \underbrace{(\text{interest to lender})}_{A \cdot r\%} + \underbrace{d}_{\text{fund deposit}}$$

A **sinking-fund accumulation schedule** tracks: deposit, interest earned by the fund, and the growing fund balance until it equals A .

Handy tabulated constants that recur in the book's examples:

$$\frac{1}{a_{\bar{6}|2}} = 0.178526, \quad a_{\bar{20}|1} = 18.045552, \quad \frac{1}{s_{\bar{10}|2}} = 0.091327$$

Why this trips people up: In amortization the borrower pays interest on a **shrinking** balance; in a sinking fund the borrower pays interest on the **full** principal the whole time (but earns interest inside the fund). They give different total costs — read which scheme the question specifies.

Chapter 4 — Bonds

A bond is a long-term promissory note: the issuer pays periodic **coupon** interest and repays the redemption value at maturity.

4.1–4.2 Terms

Symbol	Meaning
f	Face / par / maturity value — printed on the bond (usually \$100 or \$1000).
A	Redemption value — what's actually repaid. "Redeemable at 105" means $A = 1.05f$. At par, $A = f$.
b	Bond / coupon / contract rate — sets the coupon: $d = b\% \times f$ per period.
r	Yield / effective / investor's rate — the market rate the buyer actually earns. Usually $\neq b$.
n	Number of interest periods to redemption.
V_B	Purchase price = present value of the bond.

4.3 Purchase price (premium and discount)

$$V_B = p + I, \quad p = AR^{-n} \text{ (PV of redemption),} \quad I = d a_{\bar{n}|r} \text{ (PV of coupons)}$$

Algebraically equivalent, and usually faster:

$$V_B = A + (d - Ar\%) a_{\bar{n}|r}$$

Premium / discount (relative to redemption value):

$$\text{Premium} = (d - Ar\%) a_{\bar{n}|r} = A(b\% - r\%) a_{\bar{n}|r} \quad (\text{when } b > r)$$

$$\text{Discount} = (Ar\% - d) a_{\bar{n}|r} = A(r\% - b\%) a_{\bar{n}|r} \quad (\text{when } b < r)$$

When face and redemption differ (not at par), use face value f for the coupon: $\text{premium} = (fb - Ar)\% a_{\bar{n}|r}$;
 $\text{discount} = (Ar - fb)\% a_{\bar{n}|r}$.

Worked example 1. \$1000, 6% bond, redeemable in $3\frac{1}{2}$ years, bought to yield 5% compounded semi-annually, interest paid semi-annually. Per period: $r = 5/2 = 2.5\%$, $n = 3.5 \times 2 = 7$. Coupon $d = 1000 \times (6/2)\% = \30 .

(i) At par ($A = 1000$): $p = 1000(1.025)^{-7} = 1000 \times 0.841265 = \841.27 .

$I = 30 a_{\bar{7}|2.5} = 30 \times 6.349391 = \190.48 . $V_B = 841.27 + 190.48 = \1031.75 . Check via the shortcut:

$$V_B = 1000 + (30 - 25) \times 6.349391 = 1000 + 31.75 = \$1031.75. \checkmark \text{ (a **premium**, since } 6\% > 5\%.)$$

(ii) Redeemable at 105% ($A = 1050$): $p = 1050(1.025)^{-7} = \$883.33$; I is unchanged = \$190.48.

$$V_B = 883.33 + 190.48 = \$1073.81.$$

Worked example 2. \$1000, 5% bond, redeemable in $3\frac{1}{2}$ years, bought to yield **6%** semi-annually. $r = 3\%$, $n = 7$, $d = \$25$. At par: $p = 1000(1.03)^{-7} = 1000 \times 0.813092 = \$813.09\ldots \rightarrow$ this comes out **below** \$1000 (a **discount**, since $5\% < 6\%$).

4.3.1 Premium amortization / discount accumulation

Each period, the coupon differs from the yield the buyer “should” earn on the book value, so the book value drifts toward redemption value:

- **Premium bond:** book value written **down** each period by (coupon d) – (yield $\times$ book value).
- **Discount bond:** book value written **up** each period by (yield $\times$ book value) – (coupon d).

A schedule (book’s Tables 4.1 / 4.2) tracks coupon, yield interest, amortization/accumulation amount, and book value — ending exactly at A .

Why this trips people up: using the **bond rate** where the **yield rate** belongs (or vice versa). The coupon d always uses the *bond* rate; the discounting factors R^{-n} and $a_{\bar{n}|r}$ always use the *yield* rate.

Chapter 5 — Depreciation

Decline in an asset’s value over its useful life. Notation: p = new cost, s = scrap/salvage value, n = useful life (years), V = book value at time t .

5.1 Linear (straight-line) method

Equal write-off every year.

$$\text{yearly write-off} = \frac{p - s}{n}, \quad \text{monthly write-off} = \frac{p - s}{12n}$$

$$V = p - (p - s)\frac{t}{n} \quad (\text{if no scrap, } s = 0: V = p(1 - \frac{t}{n}))$$

Amount depreciated by time t : $(p - s)\frac{t}{n}$.

Worked example 1. Machine costs \$340, scrap \$75, life 5 yr. Monthly write-off
 $= \frac{340 - 75}{12(5)} = \frac{265}{60} = \4.42 . Book value at 2 yr $= 340 - 265(\frac{2}{5}) = 340 - 106 = \234 .

5.2 Sum-of-years'-digits method

Bigger write-off early. Denominator = sum of the digits $S_n = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$. Numerator for year k is $n - k + 1$ (largest first). So:

$$\text{depreciation in year } k = (p - s) \cdot \frac{n - k + 1}{S_n}$$

Worked example 3. \$340 cost, \$75 scrap, life 5. Total depreciation $= 265$, $S_5 = \frac{5(6)}{2} = 15$. Year 1:
 $265 \times \frac{5}{15} = \88.33 . Year 2: $265 \times \frac{4}{15} = \70.67 . Two-year total $= 159 \rightarrow$ book value $= 340 - 159 = \$181$.

Worked example 4. Equipment \$61050, scrap \$9000, life 3. Total $= 52050$, $S_3 = 6$. Year 1:
 $52050 \times \frac{3}{6} = \26025 (monthly \$2168.75). Year 2: $52050 \times \frac{2}{6} = \17350 (monthly \$1445.83). Book value
 end of year 2 $= 61050 - (26025 + 17350) = \17675 .

5.3 Double-declining-balance method

Rate is **twice** the linear rate, $\frac{2}{n}$, applied to the *current book value*. **Scrap value is ignored** in the calculation (depreciation could run forever).

$$\text{depreciation}_{\text{year}} = \frac{2}{n} \times (\text{book value at start of year})$$

Worked example 5. Machine \$45150.75, life 5 $\rightarrow$ rate $= \frac{2}{5} = 40\%$. Yr 1: $0.40 \times 45150.75 = 18060.30$
 $\rightarrow$ BV 27090.45. Yr 2: $0.40 \times 27090.45 = 10836.18 \rightarrow$ BV 16254.27. Yr 3: $0.40 \times 16254.27 = 6501.71 \rightarrow$ BV
 \$9752.56. (The stated \$6000 scrap never entered the arithmetic — that's the point of the method.)

5.4 Fixed-rate (constant-percentage) method

Depreciate by a fixed rate $r\%$ of book value each year, but here r is **solved from the data** (scrap not ignored):

$$V = p(1 - \frac{r}{100})^t$$

Solve for the rate using cost, scrap, and life: $s = p(1 - \frac{r}{100})^n$.

Worked example 6. Cost \$340, scrap \$75, life 5. $75 = 340(1 - \frac{r}{100})^5 \Rightarrow (1 - \frac{r}{100})^5 = \frac{75}{340} \Rightarrow r \approx 26.1\%$.

Worked example 7. New cost \$5425.50, value \$1953.18 after 2 yr.
 $1953.18 = 5425.50(1 - \frac{r}{100})^2 \Rightarrow 0.36 = (1 - \frac{r}{100})^2 \Rightarrow 0.6 = 1 - \frac{r}{100} \Rightarrow r = 40\%$. Book value at 3 yr
 $= 5425.50(0.6)^3 = \$1171.91$.

Why this trips people up: mixing up the four methods' treatment of scrap. **Linear** and **sum-of-years'-digits** use $(p - s)$. **Double-declining** ignores s entirely. **Fixed-rate** uses s only to *back out the rate*, then applies the rate to book value.

Chapter 6 — Applications of Mathematical Methods in Economics

Calculus applied to cost, revenue, profit, elasticity, and consumption. "Marginal ____" always means **the derivative** of that quantity.

6.1 Marginal cost and marginal revenue

Total revenue $R(x) = p \cdot x$, Profit $\pi(x) = R(x) - C(x)$

Marginal cost $= \frac{dC}{dx}$, Marginal revenue $= \frac{dR}{dx}$, Marginal profit $= \frac{d\pi}{dx}$

Maximize profit: set $\frac{d\pi}{dx} = 0$ (necessary), then check $\frac{d^2\pi}{dx^2} < 0$ (sufficient). Equivalent test: **MC = MR** at the optimum.

Example 1. $C = \frac{1}{2}x^2 + 3x - 20$, $R = 15x - x^2$. MC $= x + 3$; MR $= 15 - 2x$; profit $= R - C = 12x - \frac{3}{2}x^2 + 20$.

Example 2 (minimize average cost). $C(q) = 8000 - 25q + 0.02q^2$. $\frac{dC}{dq} = -25 + 0.04q = 0 \Rightarrow q = 625$ units.

Example 3 (demand $q = 36 - 3p$). Revenue $R = pq = 36p - 3p^2$. At $p = 2$: $R = 72 - 12 = \$60$. MR $= 36 - 6p = 0 \Rightarrow p = \6 . Max revenue $= 36(6) - 3(6)^2 = \$108$.

Example 4 (max profit, cubic cost). $C(x) = 5000 + 0.00002x^3 - 0.018x^2 + 7.4x$, price \$20.
 $P(x) = 12.6x - 5000 - 0.00002x^3 + 0.018x^2$. $\frac{dP}{dx} = 12.6 - 0.00006x^2 + 0.036x = 0 \Rightarrow x = 850$ (reject -250).
 $\frac{d^2P}{dx^2} = -0.00012x + 0.036 = -0.066 < 0$ at $x = 850 \rightarrow$ maximum. ✓

Example 5 (monopolist). Demand $x = 2000 - 20p$, cost $C(x) = 5x + 0.0125x^2$. Inverse demand (average revenue): $p = 100 - 0.05x$. $R(x) = 100x - 0.05x^2$; MR $= 100 - 0.1x$. MC $= 5 + 0.025x$.
 $P(x) = 95x - 0.0625x^2$; set MR = MC: $5 + 0.025x = 100 - 0.1x \Rightarrow 0.125x = 95 \Rightarrow x = 760$.
 $\frac{d^2P}{dx^2} = -0.125 < 0 \rightarrow$ max. Total profit $= 760(95) - 0.0625(760)^2 = 72200 - 36100 = \36100 .

Competitive vs. monopoly. In pure competition $\frac{dR}{dx} = p$ (price is constant, MR = price). A monopolist affects price by changing output, so MR $\neq p$; the profit-max condition is MR = MC.

6.2 Elasticity of demand and supply

Percentage change in quantity per percentage change in price:

$$e = \frac{\Delta y/y}{\Delta x/x} = \frac{\Delta y}{\Delta x} \cdot \frac{x}{y} \xrightarrow{\text{continuous}} e = \frac{dy}{dx} \cdot \frac{x}{y}$$

For demand $q = f(p)$:

$$e = \frac{dq}{dp} \cdot \frac{p}{q}$$

Example 6. $q = 100 - 5p$, at $p = 3$: $q = 85$, $\frac{dq}{dp} = -5$. $e = \frac{-5(3)}{85} = \frac{-3}{17}$. Since $|e| < 1$, demand is **relatively inelastic**.

Reading $|e|$: $|e| > 1$ elastic, $|e| < 1$ inelastic, $|e| = 1$ unit-elastic. Demand elasticity is typically negative (price up $\rightarrow$ quantity down).

6.3 Marginal propensities to consume and save

With income Y , consumption C , savings S , and $Y = C + S$:

$$\text{MPC} = \frac{dC}{dY}, \quad \text{MPS} = \frac{dS}{dY}, \quad \text{MPC} + \text{MPS} = 1$$

Example. $C(y) = 10 + \frac{3}{4}y \Rightarrow \text{MPC} = \frac{dC}{dy} = \frac{3}{4}$. Since $Y = C + S$: $S = -10 + \frac{1}{4}y \Rightarrow \text{MPS} = \frac{1}{4}$. And $\frac{3}{4} + \frac{1}{4} = 1$.

✓

Why this trips people up: forgetting that $\text{MPC} + \text{MPS} = 1$ — it's the fastest way to get one from the other, and a common check-your-work step.

Exam Tricks — where marks are actually lost

Rate & period conversion (the #1 error). “Compounded quarterly/semi-annually/monthly” means you must convert *both*: rate per period = annual rate $\div$ frequency, and n = years $\times$ frequency. A 5-year 8% quarterly problem is $r = 2$, $n = 20$ — never $r = 8$, $n = 5$.

s vs. a (accumulation vs. present value). Saving up to a future target $\rightarrow s_{\bar{n}|r}$ (future value). Paying off / valuing a stream today $\rightarrow a_{\bar{n}|r}$ (present value). Bonds and sinking funds punish the mix-up hardest.

Bond rate vs. yield rate. Coupon d = **bond** rate $\times$ face. Every discount factor (R^{-n} , $a_{\bar{n}|r}$) uses the **yield** rate. Bond rate $>$ yield $\rightarrow$ premium (price above redemption); bond rate $<$ yield $\rightarrow$ discount. If your premium/discount sign is backwards, you've swapped the rates.

Which discount world? Short-term notes $\rightarrow$ simple-interest discount (true or banker's — and *banker's is always the larger* because it charges on the maturity value, not the present value). Long-term/compound $\rightarrow p = AR^{-n}$. Don't apply AR^{-n} to a 90-day note.

360 vs. 365 days. Ordinary/banker's interest uses 360; exact uses 365. The problem's wording tells you which — read it.

Depreciation method ↔ scrap value. - Linear & sum-of-years'-digits: depreciate $(p - s)$. - Double-declining-balance: **ignore** scrap; rate = $2/n$ on book value. - Fixed-rate: use scrap only to *solve for the rate*, then apply to book value.

Sign errors in elasticity. $\frac{dq}{dp}$ is negative for normal demand, so e is negative; judge elasticity by $|e|$ vs. 1. Don't drop the sign, but don't call inelastic demand "elastic" because you forgot to compare magnitudes.

Second-order check in optimization. Setting the first derivative to zero only finds a *stationary* point. State the sufficient condition ($\pi'' < 0$ for a max) — it's often worth marks on its own.

MPC + MPS = 1. Free check and a fast way to recover the other propensity.

Rounding. Carry the table factors (s, a, R^{-n}) to their full listed precision (6 decimals in this book) until the final line; rounding early throws bond prices and schedules off by dollars.

Suggested study order

1. **Compound interest** ($A = pR^n$, present value) — the foundation of everything else.
2. **The two annuity factors** ($s_{\bar{n}|r}, a_{\bar{n}|r}$) — memorize both forms and *when* each applies.
3. **Simple interest & discounting** (true vs. banker's, promissory notes) — self-contained, quick marks.
4. **Amortization & sinking funds** — direct application of the annuity factors; practice the schedules.
5. **Bonds** — the biggest single application; drill the two equivalent formulas until automatic.
6. **Depreciation** — four methods, mechanical once you nail the scrap-value rule.
7. **Economics applications** — calculus review; marginal = derivative, optimize with first/second derivative tests.